

Characterization of the continuous, differentiating myeloid cell line (HL-60) from a patient with acute promyelocytic leukemia.

In a preliminary communication, we described the establishment of a continuous human myeloid cell line (HL-60). Here we report the detailed properties of this cell line and document its derivation from the peripheral blood leukocytes of a patient with acute promyelocytic leukemia. As characterized by light and electron microscopy, the predominant cell type in both the fresh and cultured sources is a neutrophilic promyelocyte with prominent nuclear/cytoplasmic asynchrony. Up to 10% of the cultured cells spontaneously differentiate beyond the promyelocyte stage, and the proportion of terminally differentiated cells is markedly enhanced by compounds known to stimulate differentiation of mouse (Friend) erythroleukemia cells. The HL-60 cells lack specific markers for lymphoid cells, but express surface receptors for Fc fragment and complement (C3), which have been associated with differentiated granulocytes. They exhibit phagocytic activity and responsiveness to a chemotactic stimulus commensurate with the proportion of mature cells. As characteristic of transformed cells, the HL-60 cells form colonies in semisolid medium and produce subcutaneous myeloid tumors (chloromas) in nude mice. A source of colony-stimulating activity stimulated the cloning efficiency in soft agar 5--30-fold. Despite adaptations to culture, the morphological phenotype and responsiveness to chemical induction of differentiation is essentially unchanged through at least 85 passages. Cytogenetic studies reveal aneuploidy. Metaphases with 44 chromosomes predominated in vivo and in early culture passages; however, clones with 45 or 46 chromosomes became predominant with continued passaging. The most consistent karyotypic abnormalities were the deletion of chromosomes 5, 8, and X and the addition of a marker resembling a D-group acrocentric and of a submetacentric marker, most likely an abnormal E-group chromosome. No DNA herpesvirus or RNA retrovirus was isolated in the fresh or cultured cells. The HL-60 cultured cell line provides a continuous source of human cells for studying the molecular events of myeloid differentiation and the effects of physiologic, pharmacologic, and virologic elements on this process.